Primitive leading-coefficient growth and conditional irrationality of all odd zeta values

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Endpoint formulation for the ZH manuscript and the
`GPKQBKQRXYTENSORHOPF_ODD_ZETA` project

Abstract

We formulate the precise determinant, normalization, integral, and decay endpoints needed for a conditional irrationality argument for every odd zeta value $\zeta(2k+1)$, $k \geq 1$. For the fixed family $N = 3t$, $K = 30t$, $m = 3$, let $L_{k,t}$ be the absolute leading coefficient of the primitive integer determinant polynomial of degree $27t$. We prove an explicit comparison of the unnormalized Gram determinants, implying that their logarithmic ratios are $o(t^2)$ at each fixed weight. Assuming the specified ZH weight-five decay input, the additional conjecture $\limsup_{t\to\infty} t^{-2}\log(L_{k,t}/L_{2,t}) < 9/25$ for every fixed $k \geq 3$ implies irrationality at all higher odd weights by an elementary integer-denominator argument. Apéry's theorem supplies $\zeta(3)$. The conjecture is stated equivalently with explicit eventual quantifiers. The ZH seed estimate is an assumed manuscript input, not proved in this note. A pinned Lean-source correspondence distinguishes existing conditional endpoints from the new on-paper bridge.

1 Scope and main implication

Here "all odd zeta values" means $\zeta(3), \zeta(5), \zeta(7), \ldots$; the divergent value $\zeta(1)$ is excluded. We write ZH for the supplied Zetahyperbolic manuscript [1], also called HZ in the project discussion. Only the explicitly identified statements below are used. The manuscript as a whole is not adopted as an undifferentiated axiom.

The proof has two external ZH inputs: the exact integral identification **ZH-G** in Section 3, and the integer-polynomial weight-five estimate **ZH-B5** in that section. The finite rational algebra and the normalization rules are stated and proved below. The sole additional conjecture introduced here is **PRIMITIVE_GROWTH_CONJECTURE**, Section 5.

Theorem 1.1 (Conditional all-odd irrationality)**.** *For the explicitly defined ZH family of Section 2, assume* ZH-G, ZH-B5, *and* PRIMITIVE_GROWTH_CONJECTURE. *Then*

$$\zeta(2k+1) \notin \mathbb{Q} \qquad \text{for every integer } k \geq 2.$$

Together with the separate input $\zeta(3) \notin \mathbb{Q}$, supplied by Apéry's theorem [2], this gives

$$\forall k \in \mathbb{N}, \quad k \geq 1 \implies \zeta(2k+1) \notin \mathbb{Q}.$$

The proof is given in Section 6. It establishes a conditional mathematical implication. It does not establish ZH-B5 or the primitive-growth conjecture. No finite computation or theorem name is used as evidence for either asymptotic assertion.

2 The concrete rational determinant family

2.1 Rational functional and affine pencil

Let $k \geq 1$ and write $s = 2k+1$. Put $H_a^{(s)} = \sum_{j=1}^{a} j^{-s}$. Let B_n denote the Bernoulli numbers with generating function $z/(e^z-1) = \sum_{n\geq 0} B_n z^n/n!$. On the $\mathbb{Q}$-vector space of rational functions in x whose only poles are simple poles at negative integer squares, define a $\mathbb{Q}[X]$-valued linear functional by

$$\mu_{k,X}(x^e) = \frac{(-1)^e B_{2e+2}(2e+2k+1)!}{(2k)!(2e+2)!}, \qquad e \geq 0, \tag{1}$$

$$\mu_{k,X}\left(\frac{1}{x+a^2}\right) = a^{2k}(X - H_a^{(2k+1)}) + \frac{1}{2a} - \frac{1}{2k}, \qquad a \geq 1. \tag{2}$$

Polynomial division and the unique simple-pole decomposition define $\mu_{k,X}$ on the entire stated domain. These are the precise formulas of ZH Definition 2.1; no unspecified moment functional is being chosen.

For integers $m \geq 1$ and $K > N \geq 1$, set

$$h = K - N, \qquad D_j(x) = \prod_{a=1}^{j}(x+a^2), \qquad q_{N,K,m}(x) = \frac{D_N(x)^{2m}}{D_K(x)}.$$

Each $x^{i+j}q_{N,K,m}(x)$, $0 \leq i,j < h$, lies in the rational domain. Define

$$A_{k;N,K,m}(X) = \left[\mu_{k,X}\left(x^{i+j}q_{N,K,m}(x)\right)\right]_{0\leq i,j<h}, \tag{3}$$

$$\Delta_{k;N,K,m}(X) = \det A_{k;N,K,m}(X), \qquad M_{k;N,K,m} = [X]A_{k;N,K,m}(X). \tag{4}$$

The matrix pencil has rational affine entries. The notation $[X]A$ means the coefficient of X entry by entry, equivalently its derivative with respect to X.

2.2 Primitive normalization and the fixed profile

For a nonzero $F \in \mathbb{Q}[X]$, define $\mathrm{cont}_{\mathbb{Q}}(F)$ as the unique positive rational number c such that $F/c \in \mathbb{Z}[X]$ has coefficient gcd equal to one. This normalization preserves the sign of F.

From now on the canonical family is

$$t \in \mathbb{Z}_{\geq 1}, \qquad N_t = 3t, \quad K_t = 30t, \quad m = 3, \quad h_t = 27t. \tag{5}$$

Suppressing this geometry in subscripts, put

$$\begin{aligned} \Delta_{k,t} &= \Delta_{k;3t,30t,3}, & M_{k,t} &= M_{k;3t,30t,3}, \\ c_{k,t} &= \mathrm{cont}_{\mathbb{Q}}(\Delta_{k,t}), & P_{k,t} &= \Delta_{k,t}/c_{k,t} \in \mathbb{Z}[X], \\ L_{k,t} &= |\operatorname{lc}(P_{k,t})|, & \mathscr{G}_{k,t} &= \Delta_{k,t}(\zeta(2k+1)), \end{aligned} \tag{6}$$

$$V_{k,t} = P_{k,t}(\zeta(2k+1)). \tag{7}$$

Nonvanishing of Δ is proved next. In particular $L_{k,t}$ is a positive integer. The quantity c is a *divisor*; the multiplier called `normalization.scale` in the repository is $1/c$.

3 The required ZH endpoints

3.1 ZH-A: exact degree, leading coefficient, and reweighting

Proposition 3.1 (Finite algebraic endpoint)**.** *For all $k, m \geq 1$ and $K > N \geq 1$, let $v_a = -a^2$, $N < a \leq K$, and let $\mathcal{V}_{i,a} = v_a^i$, $0 \leq i < h$. Then*

$$M_{k;N,K,m} = \mathcal{V}\ \mathrm{diag}(\gamma_{k,a})_{N<a\leq K}\,\mathcal{V}^T, \tag{8}$$

$$\gamma_{k,a} = \frac{a^{2k} D_N(-a^2)^{2m}}{D_K'(-a^2)} \neq 0. \tag{9}$$

Consequently

$$\deg \Delta_{k;N,K,m} = h, \qquad \mathrm{lc}(\Delta_{k;N,K,m}) = \det M_{k;N,K,m} \neq 0. \tag{10}$$

For every integer $d \geq 0$ at the same N, K, m,

$$\det M_{k+d;N,K,m} = \det M_{k;N,K,m} \prod_{a=N+1}^{K} a^{2d}. \tag{11}$$

Proof. The residue of $x^{i+j} D_N(x)^{2m}/D_K(x)$ at $x = -a^2$ is $(-a^2)^{i+j} D_N(-a^2)^{2m}/D_K'(-a^2)$. Only the pole terms in (2) contribute to the coefficient of X. This gives (8). The nodes are distinct and all displayed γ's are nonzero, so the Vandermonde matrix and M are nonsingular. The coefficient of X^h in the determinant of an affine $h \times h$ pencil is its slope determinant. Finally $\gamma_{k+d,a} = a^{2d}\gamma_{k,a}$, proving (11). □

This is the finite algebra used from ZH Theorem 2.5. Its alternative formula with $D_N^{2m-1}/D_{\mathrm{tail}}'$ is identical to (9), since $D_K'(-a^2) = D_N(-a^2) D_{\mathrm{tail}}'(-a^2)$.

3.2 ZH-G: exact integral identification

Define, for $y > 0$,

$$w_k(y) = \frac{2(2\pi)^{2k}}{(2k)!}\, y^{2k+1} \sum_{\ell=1}^{\infty} \ell^{2k} e^{-2\pi\ell y} > 0. \tag{12}$$

Endpoint assumption 3.2 (ZH-G)**.** For every $k \geq 1$ and every rational function R in the domain of (1)–(2),

$$\mu_{k,\zeta(2k+1)}(R) = \int_0^{\infty} R(y^2) w_k(y)\, dy. \tag{13}$$

In particular this holds for all entries of (3).

This is the selected ZH integral endpoint, equations (3.1)–(3.2). For the main theorem it would suffice to assume (13) only for the entries of the canonical family. We state the full endpoint to fix its normalization and its connection to the source.

Proposition 3.3 (ZH-P: positivity and normalization)**.** *Under ZH-G, every evaluated matrix $A_{k;N,K,m}(\zeta(2k+1))$ is positive definite. On the canonical family,*

$$\deg P_{k,t} = 27t, \qquad \mathscr{G}_{k,t} > 0, \qquad V_{k,t} > 0, \tag{14}$$

$$V_{k,t} = \frac{\mathscr{G}_{k,t}}{c_{k,t}}, \qquad L_{k,t} = \frac{|\det M_{k,t}|}{c_{k,t}}, \qquad V_{k,t} = L_{k,t} \frac{\mathscr{G}_{k,t}}{|\det M_{k,t}|}. \tag{15}$$

Proof. All moments in question are finite: w_k is bounded near zero and has an exponential tail times a polynomial; $q_{N,K,m}$ is positive on $[0,\infty)$ and has at most polynomial growth. For any nonzero real polynomial p of degree less than h,

$$\int_0^\infty p(y^2)^2 q_{N,K,m}(y^2) w_k(y)\,dy > 0.$$

This is the quadratic form of the evaluated matrix. The normalization identities now follow from (6) and Proposition 3.1. □

3.3 ZH-B5: the weight-five arithmetic seed

Endpoint assumption 3.4 (ZH-B5)**.** There is $T_0 \in \mathbb{Z}_{\geq 1}$ such that for every integer $t \geq T_0$ there exist $a_t \in \mathbb{Q}_{>0}$ and $Q_t \in \mathbb{Z}[X]$ satisfying

$$Q_t = a_t \Delta_{2,t}, \qquad 0 < Q_t(\zeta(5)) \leq \exp\left(-\frac{(30t)^2}{2500}\right). \tag{16}$$

This is precisely the form of ZH equation (4.3), page 16 of the 30-page source [1], after absorbing its positive rational normalization factors into a_t. That source attributes the arithmetic estimate to a further reference. The present note does not verify that reference or prove the estimate. ZH-B5 is retained as an explicit input.

Lemma 3.5 (Transfer of the seed to the primitive polynomial)**.** *ZH-B5 implies, for all $t \geq T_0$,*

$$0 < V_{2,t} \leq e^{-\alpha t^2}, \qquad \alpha = \frac{9}{25}. \tag{17}$$

Proof. Since $P_{2,t}$ is primitive and $Q_t = a_t c_{2,t} P_{2,t}$ has integer coefficients, $d_t = a_t c_{2,t}$ is a positive integer. Indeed an integer linear combination of the coefficients of $P_{2,t}$ equals one; multiplying it by d_t proves $d_t \in \mathbb{Z}$. Thus $d_t \geq 1$ and $V_{2,t} = Q_t(\zeta(5))/d_t \leq Q_t(\zeta(5))$. Finally $900/2500 = 9/25$. □

4 A proved comparison of the raw Gram determinants

For this section let $\mathscr{G}_k = \mathscr{G}_k(N,K,m)$ denote the determinant of the integral matrix in ZH-G, with a common geometry m, N, K at adjacent weights. The following comparison is a statement about these explicitly defined integrals. ZH-G identifies them with the rational pencil values.

Lemma 4.1 (Pointwise weight comparison)**.** *For every $k \geq 1$ and $y > 0$,*

$$\frac{1}{(2k+2)!} \leq \frac{w_{k+1}(y)}{w_k(y)} \leq (2k)!(1+2\pi y)^2. \tag{18}$$

Proof. For $r \geq 1$, $x > 0$ and $q = e^{-x}$, the Eulerian polynomial identity is

$$S_r(x) := \sum_{\ell \geq 1} \ell^r e^{-\ell x} = \frac{q E_r(q)}{(1-q)^{r+1}}, \qquad 1 \leq E_r(q) \leq r!.$$

It follows by iterating $q\,d/dq$ on $q/(1-q)$. The polynomial E_r has nonnegative Eulerian coefficients, constant coefficient one, and coefficient sum $r!$. With $r = 2k$ and $x = 2\pi y$,

$$\frac{w_{k+1}(y)}{w_k(y)} = \frac{x^2}{(r+1)(r+2)} \frac{S_{r+2}(x)}{S_r(x)}.$$

Use $1/[r!(1-e^{-x})^2] \leq S_{r+2}/S_r \leq (r+2)!/(1-e^{-x})^2$, followed by $1 \leq x/(1-e^{-x}) \leq 1+x$. □

The same Eulerian formula gives the useful envelope

$$w_k(y) \le \pi^{-1} e^{-2\pi y}(1+2\pi y)^{2k+1}.$$

It justifies boundedness near zero and all polynomially weighted tail integrability used in the determinant and radial integrals.

Lemma 4.2 (Radial moment bound)**.** *Normalize the positive density*

$$\prod_{i<j}(y_i^2-y_j^2)^2 \prod_{i=1}^{h} q_{N,K,m}(y_i^2) w_k(y_i), \qquad y_i>0, \tag{19}$$

to a probability measure. Put $R=\sum_i y_i$ and $A=2h(h+k+2mN)$. Then for every integer $j\ge 0$,

$$\mathbb{E}[(2\pi R)^j] \le (A)_j, \qquad (A)_j = A(A+1)\cdots(A+j-1), \quad (A)_0=1. \tag{20}$$

Proof. Expand the positive exponential series in each w_k and apply Tonelli. For each multi-index $(\ell_1,\dots,\ell_h)$ and each interior point of the simplex $u_i>0$, $\sum_i u_i=1$, write $y_i=ru_i$. The Jacobian contributes r^{h-1}, the squared Vandermonde contributes $r^{2h(h-1)}$, and the weight monomials contribute $r^{h(2k+1)}$. The radial factor, apart from a nonnegative constant independent of r, is

$$f(r)=r^{A_0-1}e^{-\lambda r}\prod_{i=1}^{h}\frac{D_N(r^2u_i^2)^{2m}}{D_K(r^2u_i^2)}, \quad A_0=2h(h+k), \quad \lambda=2\pi\sum_i \ell_i u_i \ge 2\pi.$$

For $g(r)=r^{A-1}e^{-\lambda r}$ the ratio is

$$\frac{f(r)}{g(r)}=\prod_{i=1}^{h}\frac{\prod_{a=1}^{N}(u_i^2+a^2/r^2)^{2m}}{\prod_{a=1}^{K}(r^2u_i^2+a^2)}.$$

This is nonincreasing in r. The normalized f law is therefore stochastically dominated by the gamma law of shape A and rate λ. For completeness, if $q=f/g$ and ϕ is increasing, then

$$2\operatorname{Cov}_g(\phi,q)=\iint(\phi(r)-\phi(s))(q(r)-q(s))\,d\nu_g(r)d\nu_g(s)\le 0,$$

where ν_g is normalized g. Dividing by $\mathbb{E}_g q$ proves the domination. All relevant moments and products are integrable; bounded truncations give the same conclusion if needed. For Z with this gamma law, $\mathbb{E}[(2\pi Z)^j]=(2\pi/\lambda)^j(A)_j\le(A)_j$. The bound is independent of the simplex point and exponential indices, so it survives their positive mixture. The case $h=1$ uses the single simplex point $u_1=1$. □

Proposition 4.3 (Explicit finite Gram comparison)**.** *For $k,m\ge 1$, $K>N\ge 1$ and $h=K-N$,*

$$((2k+2)!)^{-h} \;\le\; \frac{\mathscr{G}_{k+1}}{\mathscr{G}_k} \;\le\; ((2k)!)^h\big[3+2(h+k+2mN)\big]^{2h}. \tag{21}$$

Proof. The determinant integration identity (obtained by expanding both determinants and applying Fubini) gives

$$\mathscr{G}_k=\frac{1}{h!}\int_{(0,\infty)^h}\prod_{i<j}(y_i^2-y_j^2)^2\prod_i q_{N,K,m}(y_i^2)w_k(y_i)\,d\boldsymbol{y}.$$

Consequently $\mathscr{G}_{k+1}/\mathscr{G}_k = \mathbb{E}_k \prod_i (w_{k+1}(y_i)/w_k(y_i))$. The lower bound follows from Lemma 4.1. For the upper bound, AM–GM gives

$$\prod_i (1 + 2\pi y_i)^2 \le (1 + 2\pi R/h)^{2h}.$$

By Lemma 4.2, for $0 \le j \le 2h$ we have $\mathbb{E}(2\pi R)^j \le (A)_j \le (A + 2h)^j$. Binomial expansion therefore bounds the last expectation by $(1 + (A + 2h)/h)^{2h}$, giving (21). □

Corollary 4.4 (G-SHIFT: subquadratic raw Gram ratios)**.** *For each fixed $k \ge 1$, on the profile (5),*

$$\frac{1}{t^2} \log \frac{\mathscr{G}_{k+1,t}}{\mathscr{G}_{k,t}} \longrightarrow 0. \tag{22}$$

In particular $t^{-2} \log(\mathscr{G}_{k,t}/\mathscr{G}_{2,t}) \to 0$ for every fixed $k \ge 2$.

Proof. The upper bound in (21) becomes $((2k)!)^{27t}(90t+2k+3)^{54t}$. Its logarithm is $O_k(t \log t)$. The logarithm of the lower bound is $-27t \log((2k+2)!) = O_k(t)$. Divide by t^2. The second statement follows by a finite telescoping sum over the fixed weights $2, \dots, k-1$. □

No uniform estimate in a weight k growing with t is asserted or needed. This corollary concerns the raw Gram determinant; primitive coefficient content is accounted for separately below.

5 The primitive-growth conjecture

Conjecture 5.1 (PRIMITIVE_GROWTH_CONJECTURE)**.** For the exact primitive polynomials (6),

$$\begin{aligned} &\forall k \in \mathbb{Z}_{\ge 3}\ \exists \delta_k > 0\ \exists T_k \in \mathbb{Z}_{\ge 1}\ \forall t \in \mathbb{Z}_{\ge 1}, \quad t \ge T_k \implies \\ &\qquad L_{k,t} \le L_{2,t} \exp((\alpha - \delta_k) t^2), \qquad \alpha = \frac{9}{25}. \end{aligned} \tag{23}$$

The index k is quantified *before* its margin δ_k and cutoff T_k. Neither a common margin nor a common cutoff is required. Equivalently, with limit superior understood in the extended real line,

$$\forall k \ge 3, \qquad \limsup_{t \to \infty} \frac{1}{t^2} \log \frac{L_{k,t}}{L_{2,t}} < \frac{9}{25}. \tag{24}$$

Indeed an eventual bound (23) bounds the limit superior by $\alpha - \delta_k$, and a limit superior strictly below α permits an intermediate real eventual upper bound. Formula (23) is the preferred formal specification; it avoids conventions for a real-valued limit superior of a sequence not known to be bounded above.

5.1 An equivalent statement about rational content

Set

$$W_t = \prod_{a=3t+1}^{30t} a^2 \ge 1. \tag{25}$$

From ZH-A and (15), for each $k \ge 2$,

$$\frac{|\det M_{k,t}|}{|\det M_{2,t}|} = W_t^{k-2}, \tag{26}$$

$$\frac{c_{k,t}}{c_{2,t}} = W_t^{k-2} \frac{L_{2,t}}{L_{k,t}}. \tag{27}$$

Since

$$0 \le \log W_t \le 54t \log(30t) = o(t^2), \tag{28}$$

Conjecture 5.1 is equivalent to

$$\forall k \ge 3, \qquad \liminf_{t\to\infty} \frac{1}{t^2} \log \frac{c_{k,t}}{c_{2,t}} > -\frac{9}{25}. \tag{29}$$

Here too the limit inferior is extended-real. The content sign is essential: $V = \mathscr{G}/c$, whereas the repository's `contentLog` is $\log(1/c) = -\log c$.

5.2 What the conjecture asks for arithmetically

If $d > 0$ clears the coefficient denominators of $\Delta_{k,t}$ and $d\Delta_{k,t}(X) = \sum_{j=0}^{27t} a_j X^j$, then

$$L_{k,t} = \frac{|a_{27t}|}{\gcd(|a_0|, \dots, |a_{27t}|)}. \tag{30}$$

This value is independent of d. Thus the conjecture compares primitive integer coefficient vectors of explicitly defined rational determinants; its statement contains no irrationality assumption and no evaluation at an unknown zeta value. Exact finite samples do not establish its eventual bound. A rational change of polynomial basis multiplies the determinant by a nonzero rational scalar and preserves $L_{k,t}$, so such a basis change alone cannot improve the conjectured primitive growth rate.

The stronger family of adjacent conditions $\limsup t^{-2} \log(L_{j+1,t}/L_{j,t}) \le 0$ for every fixed $j \ge 2$ would imply the conjecture by finite telescoping. We require only the direct comparison with weight five in (23).

6 From the endpoints to irrationality

Lemma 6.1 (Exact cross-weight value identity)**.** *For every $k \ge 2$ and $t \ge 1$, under ZH-G,*

$$\frac{V_{k,t}}{V_{2,t}} = \frac{\mathscr{G}_{k,t}}{\mathscr{G}_{2,t}} \frac{L_{k,t}}{L_{2,t}} W_t^{-(k-2)}. \tag{31}$$

Proof. Divide $V = L\mathscr{G}/|\det M|$ at weight k by the same identity at weight two, and apply (26). All factors are strictly positive. □

Proposition 6.2 (Primitive quadratic decay)**.** *Assume ZH-G and ZH-B5. If (23) holds at a fixed $k \ge 3$ with margin $\delta_k > 0$, then for all sufficiently large integers t,*

$$0 < V_{k,t} \le \exp\left(-\frac{\delta_k}{2} t^2\right). \tag{32}$$

At $k = 2$, the seed gives $0 < V_{2,t} \le e^{-\alpha t^2}$ eventually.

Proof. Corollary 4.4 gives $\mathscr{G}_{k,t}/\mathscr{G}_{2,t} \le \exp((\delta_k/2)t^2)$ eventually. Take a common cutoff for this estimate, the seed, and (23). Then (31) and $W_t^{-(k-2)} \le 1$ give

$$0 < V_{k,t} \le e^{-\alpha t^2} e^{(\delta_k/2)t^2} e^{(\alpha-\delta_k)t^2} = e^{-(\delta_k/2)t^2}.$$

□

Lemma 6.3 (INT: integer-polynomial denominator transfer)**.** *Let $\xi \in \mathbb{R}$, and suppose $P_t \in \mathbb{Z}[X]$ have $\deg P_t \le Dt$ for some fixed integer $D \ge 0$. If for some $\eta > 0$ and all sufficiently large positive integers t, $0 < P_t(\xi) \le e^{-\eta t^2}$, then $\xi \notin \mathbb{Q}$.*

Proof. If $\xi = a/b$, with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$, then $b^{Dt}P_t(a/b)$ is a positive integer. Hence for all sufficiently large t,

$$1 \le b^{Dt}P_t(a/b) \le \exp(Dt \log b - \eta t^2) < 1,$$

a contradiction. The final inequality follows once $t > D \log b/\eta$. □

Proof of Theorem 1.1. For $k = 2$, apply Lemma 6.3 with $P_t = P_{2,t}$, $D = 27$ and $\eta = \alpha$. For any fixed $k \ge 3$, use Proposition 6.2 and the same lemma with $\eta = \delta_k/2$. This proves every odd weight at least five. Apéry's theorem supplies the remaining $k = 1$ case independently. □

The qualitative argument uses no root-separation estimate, zero-free disk, effective irrationality measure, or uniformity in k. The number 27 is the dimension difference $30 - 3$ in the chosen family. The threshold $9/25$ is the seed rate $30^2/2500$, not a universal constant attached to odd zeta values.

7 The concrete Fermi realization of the project object

The analytical connection to the named `GPKQBKQRXYTENSORHOPF` object can be made precise for its Fermi multiplication-model component. Define

$$d\sigma(x) = \frac{2\,dx}{e^{2\pi\sqrt{x}} + 1}, \qquad \mathcal{H} = L^2((0,\infty), \sigma), \qquad (Tf)(x) = xf(x)$$

with the natural multiplication-operator domain. For $x > 0$, put

$$\omega_k(x) = \frac{(2\pi)^{2k}x^k}{2(2k)!}(e^{2\pi\sqrt{x}} + 1)\sum_{\ell \ge 1} \ell^{2k}e^{-2\pi\ell\sqrt{x}}. \tag{33}$$

Changing variables $x = y^2$ gives the exact measure identity

$$\omega_k(x)\,d\sigma(x) = \frac{w_k(\sqrt{x})}{2\sqrt{x}}\,dx.$$

Thus, for $0 \le i < h$,

$$u_{k,i}(x) = x^i\sqrt{q_{N,K,m}(x)\omega_k(x)} \in \mathcal{H}, \qquad A_{k;N,K,m}(\zeta(2k+1))_{ij} = \langle u_{k,i}, u_{k,j}\rangle_{\mathcal{H}}. \tag{34}$$

Membership follows from the finite moments already used in Proposition 3.3. This is an explicit Gram realization of the same evaluated ZH pencil in one Hilbert space.

The reweighting factor ω_k is necessary. In particular $\sigma((0,\infty)) = 1/12$, whereas $\int \omega_k\,d\sigma = \mu_{k,X}(1) = (2k+1)/12$. This section does not identify every historical tensor or Hopf-algebra component of the named object with the ZH pencil. Nor does a Hilbert-space Gram realization supply a bound on integer coefficient content. The arithmetic quantity in Conjecture 5.1 remains the exact primitive leading coefficient (30).

8 Lean endpoint correspondence and evidence boundary

The following mapping uses the pinned repository [3], commit

`211b362c16ff6ef73e9a5f2f70c5cbb9a444cf80`.

The repository uses $n \in \mathbb{N}$, with $t = n + 1$, and its `affineFamily k n` is exactly $\langle k, 3, 3(n + 1), 30(n + 1)\rangle$. The source statements below must be read with their hypotheses; a conditional transfer theorem does not prove its decay premise.

Endpoint	Declaration (under `ManuscriptChain`)	Exact role
ZH-A	`PoleDiagonalization.rationalDerivativeMatrix_det_reweight_add`	Finite derivative-determinant reweighting.
ZH-G / ZH-P	`REALIZATION`; `GRAM_POSITIVE`	Integral realization and positivity for admissible parameters.
Primitive family	`GeneralIrrationality.primitivePolynomial_natDegree`; `GeneralIrrationality.primitivePolynomial_eval_pos`	Exact degree and positive primitive evaluation.
Content identity	`ContentGramCancellation.primitive_eval_eq_scale_mul_gram`	$V = (1/c)\mathscr{G}$.
Leading identity	`LeadingCoefficientGap.leadingSize_eq_scale_mul_abs_det`	$L = (1/c)\lvert\det M\rvert$.
Rate transfer	`LeadingCoefficientGap.normalizedQuadraticDecay_of_gap`	Decay follows from two explicit eventual rate bounds and a strict gap.
INT	`RealTransfer.irrational_of_linear_degree_quadratic_decay`	Nonzero integer-polynomial values and decay imply irrationality.
Fixed weight	`GeneralIrrationality.irrational_riemannZeta_of_normalized_decay`	Returns irrationality of Mathlib's actual Riemann zeta real part.
All weights	`GeneralIrrationality.all_odd_zeta_irrational_of_affineFamilyDecay`	Requires affine-family decay at *every* $k > 0$.

More precisely, in the namespace `ManuscriptChain.GeneralIrrationality`, the existing final declaration is

```
all_odd_zeta_irrational_of_affineFamilyDecay
  (h : ∀ k : ℕ, 0 < k → AffineFamilyDecay k) :
  ∀ k : ℕ, 0 < k →
    Irrational (riemannZeta ((2 * k + 1 : ℕ) : ℂ)).re
```

For the present route, the fixed-weight theorem should be applied for $k \geq 2$ and then combined with the separate $\zeta(3)$ input. There is no need to assert decay of the canonical family at $k = 1$ merely to use the universal wrapper.

The repository's `NormalizedQuadraticDecay` uses $e^{-cK_t^2}$, whereas this note uses $e^{-\eta t^2}$. The conversion is $c = \eta/900$. The seed therefore has $c = 1/2500$, and the conclusion (32) has $c = \delta_k/1800$.

Obligation	Status in this preprint
Finite rational structure	Derived in Proposition 3.1; corresponding source endpoints exist.
ZH-G	Explicitly selected integral input; corresponding concrete repository realization exists.
ZH-B5	Explicit assumed manuscript estimate; no proof or numerical certificate supplied here.
G-SHIFT	Proved on paper in Section 4; no new Lean certification claimed.
Primitive-growth bound	Conjecture 5.1, still unproved.
Conjecture-to-decay assembly	Proved on paper in Section 6; its new Lean assembly remains to be built and checked.
Odd-zeta conclusion	Conditional on the stated inputs and conjecture.

The inspected GitHub Linux run [4] at the pinned commit compiled the relevant modules. Its printed dependencies for the final conditional endpoint are `propext`, `Classical.choice`, and `Quot.sound`. The complete 238-module run nevertheless failed later in `LeanPaper.Section5RationalIndependence`, before the final project-wide axiom audit. These observations neither certify the new proof in this note nor constitute native Windows compiler evidence. No Lean compilation was performed while preparing this preprint.

References

[1] Prince Anand, *Spectral Theory and Diophantine Bounds for Hankel Determinants of Odd Zeta Values*, supplied manuscript dated 21 October 2026, 30 PDF pages, filename `Zetahyperbolic (1).pdf`. Used items: Definition 2.1, equations (2.2)–(2.3), (2.8)–(2.11), (3.1)–(3.2), and (4.2)–(4.3). SHA-256: `11b822475775417ab36a63dd7ffd186e58970a358716e89968bd01a0ea821839`. This identifies the inspected version; it does not assert identity with other manuscript versions associated with the repository.

[2] Roger Apéry, *Irrationalité de* $\zeta 2$ *et* $\zeta 3$, Journées Arithmétiques de Luminy, Astérisque **61** (1979), 11–13. https://www.numdam.org/item/AST_1979__61__11_0/.

[3] `randogelist/zetahyperbolic-axiom`, pinned source snapshot inspected 26 October 2026. Repository at commit 211b362c16ff. Principal files: `MAIN_CHAIN_ASSEMBLED.lean`, `MAIN_CHAIN_IRRATIONALITY.lean`, `Theorem_2_5_Weight_Change.lean`, `Theorem_4_1_Content_Gram_Cancellation.lean`, `Theorem_4_1_Leading_Coefficient_Gap.lean`, and `Theorem_4_1_Integer_Polynomial_Irrationality.lean`.

[4] GitHub Actions, *Leaned Paper — compile and axiom audit*, run `40281351796`, job `121435142988`, 26 October 2026. Pinned-commit compilation log. Overall conclusion: failure after the relevant earlier module builds; complete project-wide audit not reached.